Great Expectations Task

1. Install Great Expectations Library

```
!pip install great_expectations
```

```
Requirement already satisfied: great_expectations in /usr/local/lib/python3.12/dist-packages (1.16.0)
Requirement already satisfied: altair<7.0.0,>=5.0.0 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (5.5.0)
Requirement already satisfied: cryptography>=3.2 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (43.0.3)
Requirement already satisfied: jinja2>=3 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (3.1.6)
Requirement already satisfied: jsonschema>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (4.26.0)
Requirement already satisfied: marshmallow<4.0.0,>=3.7.1 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (3
Requirement already satisfied: mistune>=0.8.4 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (3.2.0)
Requirement already satisfied: numpy>=1.22.4 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (2.0.2)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from great_expectations) (26.0)
Requirement already satisfied: pandas>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (2.2.2)
Requirement already satisfied: pydantic>=1.10.7 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (2.12.3)
Requirement already satisfied: pyparsing!=3.2.4,>=2.4 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (3.3.
Requirement already satisfied: python-dateutil>=2.8.1 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (2.9.
Requirement already satisfied: requests>=2.20 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (2.32.4)
Requirement already satisfied: ruamel.yaml>=0.16 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (0.19.1)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (1.16.3)
Requirement already satisfied: tqdm>=4.59.0 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (4.67.3)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (4.
Requirement already satisfied: tzlocal>=1.2 in /usr/local/lib/python3.12/dist-packages (from great_expectations) (5.3.1)
Requirement already satisfied: narwhals>=1.14.2 in /usr/local/lib/python3.12/dist-packages (from altair<7.0.0,>=5.0.0->great_exp
Requirement already satisfied: cffi>=1.12 in /usr/local/lib/python3.12/dist-packages (from cryptography>=3.2->great_expectations
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2>=3->great_expectations) (
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.5.1->great_expectati
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema>
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.5.1->great_exp
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.5.1->great_expecta
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.3.0->great_expectations)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.3.0->great_expectations
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic>=1.10.7->great_e
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic>=1.10.7->great_ex
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic>=1.10.7->great
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.1->great_expectati
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.20->great_e
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.20->great_expectations)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.20->great_expecta
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.20->great_expecta
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi>=1.12->cryptography>=3.2->great_e
```

2. Import Necessary Libraries

```
import pandas as pd
import great_expectations as gx
```

3. Load Labels.csv

Download and upload the [Labels.csv]() into this notebook, and then load the file.

```
df = pd.read_csv("/content/Labels.csv")
```

```
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is depreca
  return datetime.utcnow().replace(tzinfo=utc)
```

4. Preview the Dataset

```
df.head()
```

	Timestamp	Car1_Location_X	Car1_Location_Y	Car1_Location_Z	Car2_Location_X	Car2_Location_Y	Car2_Location_Z	Occluded_Im
0	1736796157	-51.402977	143	0.596902	-59.320270	140	0.596902	A
1	1736796167	-53.819637	143	0.596902	-59.196568	140	0.596902	A
2	1736796178	-50.239144	143	0.596902	-56.744479	140	0.596902	A
3	1736796188	-53.707220	143	0.596902	-57.309380	140	0.596902	A
4	1736796198	-52.053721	143	0.596902	-59.545897	140	0.596902	A

```
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecat
  return datetime.utcnow().replace(tzinfo=utc)
```

Next steps: Generate code with df New interactive sheet

5. Set Up Great Expectations Context and Data Source

```
context = gx.get_context()
data_source = context.data_sources.add_pandas("pandas")
data_asset = data_source.add_dataframe_asset(name="pd dataframe asset")
```

```
INFO:great_expectations.data_context.types.base:Created temporary directory '/tmp/tmpm5bzh109' for ephemeral docs site
```

6. Define and Create a Data Batch

```
batch_definition = data_asset.add_batch_definition_whole_dataframe("batch definition")
batch = batch_definition.get_batch(batch_parameters={"dataframe": df})
```

```
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecat
  return datetime.utcnow().replace(tzinfo=utc)
```

7. Define Three Expectations for Column Values

Using this link, choose three expectation functions and apply them to the labels dataset in a relevant manner.

You should replace the 'ExpectColumnValuesToBeBetween' function with other functions you select from the link.

You can also check the format/parameters required of each function when you click "See more" on the function.

```
## Original Function
expectation = gx.expectations.ExpectColumnValuesToBeBetween(
    column="column", min_value=0, max_value=20
)

## Example Function

## This function only requires a column parameter, and not a max or min value
expectation = gx.expectations.ExpectColumnValuesToBeUnique(
    column="column"
)
```

Expectation 1

```
expectation1 = gx.expectations.ExpectColumnValuesToBeUnique(
    column="Occluded_Image_view"
)
```

Validate Data Against Expectation 1

```
validation_result1 = batch.validate(expectation1)
print(validation_result1)
```

```
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is depreca
  return datetime.utcnow().replace(tzinfo=utc)
Calculating Metrics: 100%                                    10/10 [00:00<00:00, 475.76it/s]
{
  "success": false,
  "expectation_config": {
    "type": "expect_column_values_to_be_unique",
    "kwargs": {
      "batch_id": "pandas-pd dataframe asset",
      "column": "Occluded_Image_view"
    },
    "meta": {},
    "severity": "critical"
  },
  "result": {
    "element_count": 121,
    "unexpected_count": 2,
    "unexpected_percent": 1.6528925619834711,
    "partial_unexpected_list": [
      "A_085.png",
      "A_085.png"
    ],
    "missing_count": 0,
    "missing_percent": 0.0,
    "unexpected_percent_total": 1.6528925619834711,
    "unexpected_percent_nonmissing": 1.6528925619834711,
    "partial_unexpected_counts": [
      {
        "value": "A_085.png",
        "count": 2
      }
    ],
    "partial_unexpected_index_list": [
      84,
      85
    ]
  },
  "meta": {},
  "exception_info": {
    "raised_exception": false,
    "exception_traceback": null,
    "exception_message": null
  }
}
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is depreca
  return datetime.utcnow().replace(tzinfo=utc)
```

Expectation 2

```
expectation2 = gx.expectations.ExpectColumnValuesToBeBetween(
    column="Timestamp",
    min_value=0.0,
    strict_min=True
)
```

Validate Data Against Expectation 2

```
validation_result2 = batch.validate(expectation2)
print(validation_result2)
```

```
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecat
  return datetime.utcnow().replace(tzinfo=utc)
Calculating Metrics: 100%                                                    10/10 [00:00<00:00, 294.43it/s]
{
  "success": true,
  "expectation_config": {
    "type": "expect_column_values_to_be_between",
    "kwargs": {
      "batch_id": "pandas-pd dataframe asset",
      "column": "Timestamp",
      "min_value": 0.0,
      "strict_min": true
    },
    "meta": {},
    "severity": "critical"
  },
  "result": {
    "element_count": 121,
    "unexpected_count": 0,
    "unexpected_percent": 0.0,
    "partial_unexpected_list": [],
    "missing_count": 0,
    "missing_percent": 0.0,
    "unexpected_percent_total": 0.0,
    "unexpected_percent_nonmissing": 0.0,
    "partial_unexpected_counts": [],
    "partial_unexpected_index_list": []
  },
  "meta": {},
  "exception_info": {
    "raised_exception": false,
    "exception_traceback": null,
    "exception_message": null
  }
}
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecat
  return datetime.utcnow().replace(tzinfo=utc)
```

Expectation 3

```
expectation3 = gx.expectations.ExpectColumnValuesToNotBeNull(
    column="Car1_Location_X"
)
```

Validate Data Against Expectation 3

```
validation_result3 = batch.validate(expectation3)
print(validation_result3)
```

```
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecat
  return datetime.utcnow().replace(tzinfo=utc)
Calculating Metrics: 100%                                                    8/8 [00:00<00:00, 148.29it/s]
{
  "success": true,
```